

Emotion-Aware Personalized Co-Speech Motion Generation

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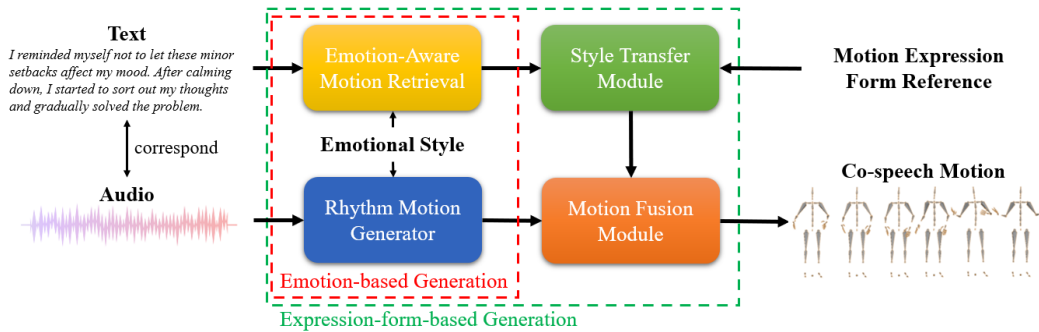


Figure 1: There is our emotion-aware personalized co-speech motion generation system

ABSTRACT

Co-speech motion generation is a challenging task requiring alignment between motion, speech rhythm, and semantics. To improve adaptability, style control mechanisms, including emotion perception and personalized motion generation, are needed. We propose a hybrid emotion-aware personalized co-speech motion generation system, consisting of an emotion-aware module based on LLM and a rhythm motion generation module based on diffusion, ensuring consistency between motion and speech emotion. Additionally, we introduce a style transfer module to adapt the generated motions to the speaker's style. Experimental results show the system generates stylized and emotionally consistent motions, enhancing the realism and stylization of virtual humans.

Index Terms: Co-speech motion generation, Style transfer, Large Language Model

1 INTRODUCTION

Intelligent virtual humans generate behaviors, like facial and body movements, influenced by external factors such as conversation, with applications in VR. Co-speech motions enhance verbal expression, convey emotions, and increase realism, while personalized motions reflect unique character designs, making virtual humans more natural. However, challenges remain in generating motions that align with speech rhythm, semantics, and emotions, while supporting personalization. To address this, we propose a hybrid sys-

tem combining LLM, diffusion, and style transfer, enabling personalized motions aligned with emotions.

There are our contributions in this work:

- (1) We propose a novel hybrid system for emotion-aware personalized co-speech motion generation.
- (2) We utilize an LLM to perform emotion-aware semantic motion retrieval within the constructed ethogram, enhancing the emotional expressiveness of the motions.
- (3) We achieve control personalized motion expression form through the application of style transfer.

2 SYSTEM OVERVIEW

To achieve co-speech motion generation that aligns with both speech rhythm and semantics, we propose a hybrid system based on Zhang's approach[3], comprising four modules: an emotion-aware motion retrieval module, a style transfer module, a rhythm motion generator, and a motion fusion module. Motion style is defined as the combination of emotion and motion expression form. Emotional control is managed by assigning emotional probability labels to semantic motions, which guide the LLM-based selection, while the style transfer module handles motion style control by mapping the desired styles to motions.

2.1 Emotion-Aware Motion Retrieval

We have developed a co-speech ethogram (available at <https://github.com/gesture-label/ethogram>) to summarize motions a digital human may perform during speech. The ethogram has three layers: the intent layer, provides a high-level abstraction of the motivations; the sub-intent layer, further refining these motivations; and the action layer, mapping intents to specific motions. This structure supports LLM classification. We used prompt engineering with LLMs, employing chain-of-thought and self-reflective strategies to select motions matching the text. The LLM is fine-tuned with motion prompts and emotional style-based probability

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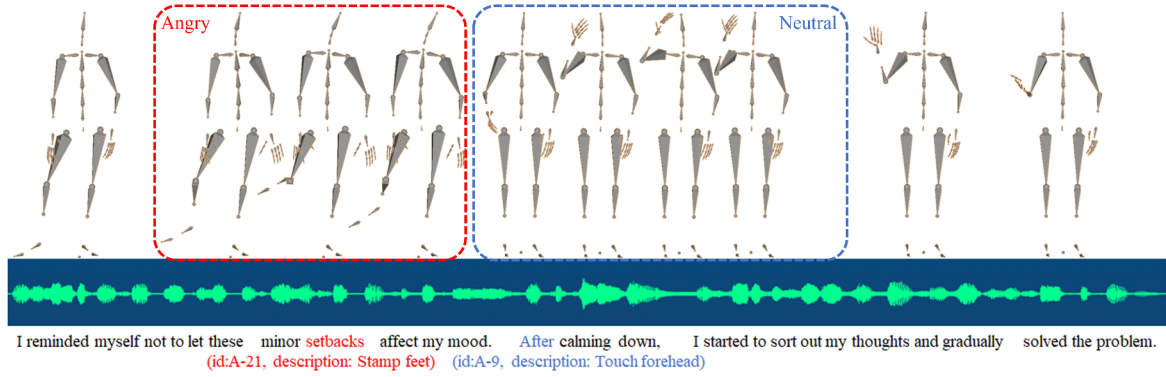


Figure 2: The figure showcases results of emotion-aware motion generation. In the red box, anger is vividly expressed through tense, explosive actions, with exaggerated joint angles and rapid movements. In contrast, the blue box depicts neutral motions, featuring relaxed, smooth, and moderate actions, reflecting calmness and tranquility.

values, enabling the selection of semantic motions aligned with the desired emotional style.

2.2 Style Transfer Module

To overcome the limitation of universal motions in the co-speech ethogram, we introduce a style transfer method that reduces its size while ensuring flexibility and personalization. Inspired by Aberman[1], our style transfer module includes two encoders, a decoder, and a discriminator. The encoders extract style and content motion encodings, while the decoder uses AdaIN to embed style into content and generate new motions. The discriminator evaluates the realism of the generated motions and ensures effective style transfer. Following these operations, the motion expression forms are effectively mapped to the selected semantic motions.

2.3 Rhythm Motion Generator

To better enhance the influence of speech rhythm on motions and ensure good diversity in the generated motions, we choose to use diffusion for generating rhythmic motions[2]. We also use WavLM to extract speech features. WavLM extracts emotional features from speech audio, which are combined with other inputs as auxiliary information. Starting with noisy motions, the diffusion model progressively denoises them. Using Cross-Local Attention and self-attention, features are fused to generate the motion sequence, enabling effective emotional style control for rhythmic motion generation.

2.4 Motion Fusion Module

We obtain effective semantic and rhythm motions, which are then fused. To preserve alignment between rhythmic motions and speech rhythm, the fusion timing is based on the speech beat nearest to the LLM-approved keyword, using it as the peak for inserting semantic motions. A dynamic transition ensures a smooth blend between the two motions, maintaining rhythm consistency with the speech.

3 RESULT AND DISCUSSION

The results of our emotion-aware personalized motion generation method, shown in the Figure 2, demonstrate the virtual human’s motion generation across different motion styles, while accurately matching the rhythm and content of the speech. Each motion segment is closely linked with the corresponding speech content, such as “stomping” and “touching the forehead.” We used angry and neutral speech audio to drive motion generation. As a result, each motion segment exhibits distinct emotions and expression forms, closely aligned with our control. Through our user experiments,

most participants agreed that our method achieved favorable results, appropriately conveying the semantics and emotions of the speech.

Our system takes speech text, corresponding audio, emotional style, and personalized motion expression form references as inputs. First, emotion-aware motion retrieval is performed based on the text and emotion to obtain semantically relevant motions. Then, style transfer is applied to map personalized expression forms onto these semantic motions, incorporating user preferences. Rhythm motions are generated based on the speech audio and its emotional style, ensuring synchronization with speech rhythm and tone. Finally, the two types of motions are fused, ensuring the virtual human’s motions align with both the speech content and emotional/contextual variations.

These results validate the effectiveness of our method in coordinating speech and motion generation, ensuring high-quality motion performance and accurate motion style representation. As a result, it enhances the lifelike quality of virtual humans, improving the sense of immersion for the VR audience.

4 CONCLUSION

Our method offers significant advantages in generating motions driven by speech content and style. By integrating a style transfer module, the system captures speech content while generating diverse motions based on emotions and personalized expression forms. Experimental results and user tests demonstrate that our method significantly enhances the expressiveness and interactivity of virtual humans.

5 ACKNOWLEDGEMENT

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